

PORVOO EMASALO, Finland

WMO: 029910

Lat: 60.204N

Lon: 25.626E

Elev: 27

StdP: 101.00

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.8	-15.7	-21.2	0.6	-18.6	-17.9	0.8	-15.0	17.8	0.5	16.2	0.4	5.4	40	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.1	24.6	19.9	23.1	19.1	21.6	17.9	21.1	23.7	19.8	22.5	18.6	21.0	5.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	14.7	23.2	18.7	13.6	21.8	17.5	12.6	20.2	61.1	23.8	56.9	22.6	52.7	21.0	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	15.4	14.0	DB	-20.0	26.5	4.3	2.8	-23.1	28.5	-25.6	30.2	-27.9	31.8	-31.0	33.8	(4)
(5)			WB	-20.3	21.1	4.2	1.5	-23.3	22.1	-25.8	23.0	-28.1	23.8	-31.2	24.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-3.9	-4.4	-1.6	3.6	9.7	14.0	17.9	17.1	12.9	7.2	3.1	0.1		
	DBStd	8.53	5.53	5.53	3.90	2.66	3.67	2.72	2.98	2.48	2.73	3.29	3.77	4.35		
	HDD10.0	2058	430	404	358	194	49	2	0	0	8	97	208	307		
	HDD18.3	4416	688	638	617	443	267	133	44	53	163	347	458	566		
	CDD10.0	736	0	0	0	1	41	122	246	222	95	9	0	0		
	CDD18.3	51	0	0	0	0	1	3	32	16	0	0	0	0		
	CDH23.3	111	0	0	0	0	1	3	89	18	0	0	0	0		
	CDH26.7	6	0	0	0	0	0	0	6	0	0	0	0	0		
(14) Wind	WSAvg	7.0	7.7	6.8	7.0	6.3	6.1	6.4	5.8	6.4	7.0	7.6	8.4	8.9		
(15) Precipitation	PrecAvg	672	57	44	36	35	39	65	64	73	62	70	73	58		
	PrecMax	865	113	105	67	72	74	136	148	146	155	178	187	147		
	PrecMin	542	8	3	7	7	4	17	1	4	13	20	10	14		
	PrecStd	91	24	26	15	19	22	32	36	35	39	36	35	28		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	4.2	6.9	14.4	21.6	23.1	27.3	25.1	19.4	14.0	9.6	7.5		
		MCWB	4.2	2.3	3.7	8.7	13.6	17.3	21.0	20.9	16.3	12.3	8.8	6.7		
	2%	DB	4.0	2.5	4.9	10.7	19.2	20.9	25.5	23.5	18.2	13.1	9.0	6.5		
		MCWB	3.6	1.6	2.6	6.6	13.9	16.2	20.4	19.7	15.8	12.1	8.4	5.7		
	5%	DB	3.3	1.7	3.5	8.8	17.1	19.5	24.1	22.1	17.2	12.3	8.2	5.7		
		MCWB	2.8	1.3	1.9	5.4	12.7	15.3	19.7	18.5	15.2	11.4	7.6	5.0		
	10%	DB	2.4	1.2	2.5	7.2	15.2	18.1	22.7	20.8	16.4	11.5	7.5	4.8		
		MCWB	1.9	0.8	1.4	4.7	11.6	14.3	19.1	17.8	14.8	10.6	6.9	4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	2.7	4.2	9.1	16.2	19.0	22.5	21.9	17.4	13.1	9.2	7.0		
		MCDB	4.7	3.7	6.0	12.7	19.0	21.3	25.5	24.5	18.9	13.4	9.3	7.4		
	2%	WB	3.7	1.9	3.1	7.2	14.5	17.3	21.6	20.5	16.5	12.4	8.5	5.9		
		MCDB	4.0	2.3	4.3	10.0	17.5	20.1	24.4	22.8	17.6	12.8	8.7	6.3		
	5%	WB	2.9	1.5	2.3	6.0	13.4	16.0	20.7	19.3	15.8	11.6	7.7	5.1		
		MCDB	3.2	1.8	3.2	8.0	16.5	18.5	23.4	21.5	16.8	12.1	8.1	5.6		
	10%	WB	2.0	0.9	1.7	5.1	11.9	15.0	19.4	18.3	15.1	10.8	7.0	4.3		
		MCDB	2.4	1.2	2.3	6.8	15.2	17.5	22.3	20.3	16.2	11.4	7.4	4.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.0	4.6	4.8	5.4	5.1	5.1	4.6	4.1	3.7	3.0	3.3		
		MCDBR	2.8	2.6	4.6	7.0	7.7	7.0	6.7	5.7	4.7	3.4	2.7	2.9		
	5% WB	MCWBR	3.1	2.4	3.3	4.1	4.4	3.9	3.7	3.4	3.2	3.0	2.9	3.0		
		MCWBR	2.8	2.6	3.9	5.9	7.2	5.9	5.8	5.0	4.0	3.1	2.6	2.7		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.310	0.333	0.363	0.362	0.358	0.380	0.373	0.347	0.333	0.306	N/A		
		taud	2.140	2.181	2.233	2.325	2.410	2.468	2.430	2.446	2.508	2.394	2.213	N/A		
	Edn at Noon		474	670	786	834	865	874	842	818	782	654	436	N/A		
		Edh at Noon	43	73	95	105	103	99	101	93	74	60	41	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.25	0.82	2.31	3.85	5.25	5.46	5.29	4.09	2.49	1.08	0.30	0.15		
	RadStd		0.04	0.16	0.30	0.40	0.57	0.42	0.48	0.42	0.32	0.17	0.05	0.02		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (61 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	-148	-163	N/A	N/A

Nomenclature: See separate page